Midterm Review

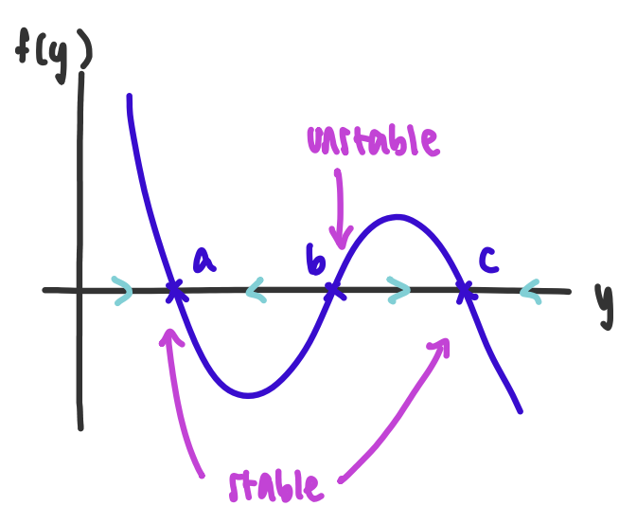
Tuesday, October 21, 2025 4:49 PM

$e^{i\phi} = \cos\phi + i\sin\phi$

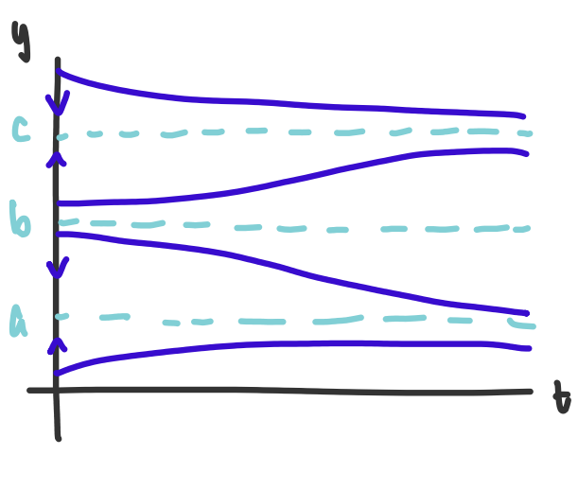
Newton's law of cooling: $u' = -k(u - T_0)$

Autonomous: $\frac{dy}{dt} = f(y)$ (no t term)

Equilibrium solns: $f(y) = 0$ (critical, fixed, stationary points)



phase portrait



y vs. t

Linear differential equation: n^{th} order ODE is linear if it can be written in the form

$$a_0(t)y^{(n)} + a_1(t)y^{(n-1)} + \dots + a_n(t)y = \underline{g(t)}$$

↑ if $=0$, then homogeneous

Separable differential equation: $\frac{dy}{dx} = p(x)q(y)$

$$\frac{dy}{q(y)} = p(x)\,dx$$

$$\int \frac{dy}{q(y)} = \int p(x)\,dx$$

...

(t,y) direction field

Autonomous: slope is independent of t

Linear: slope varies uniformly with y (proportional changes)

Homogeneous: slope is 0 at (0,0)

Separable: slope varies as a function of t for a fixed y,

slope varies as a function of y for a fixed t

autonomous ⟹ separable

Linear autonomous system can have 0, 1 or ∞ equilibria.

Method of Integrating Factors

$$\frac{dy}{dt} + p(t)y = g(t)$$

IF: $\mu(t) = e^{\int p(t)\,dt}$

$$[\mu(t)y]' = \mu(t)g(t)$$

$$\mu(t)y = \int \mu(t)g(t)\,dt + C$$

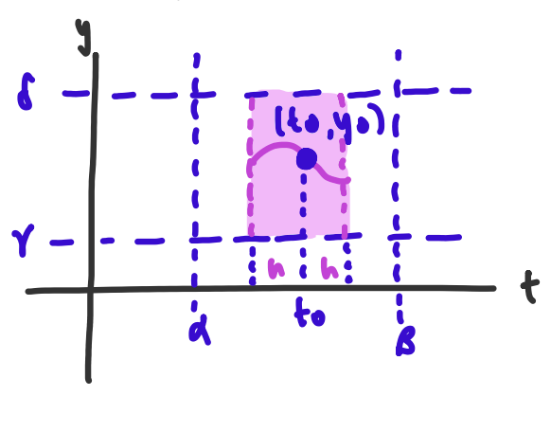
$$y = \frac{1}{\mu(t)}\left[\int \mu(t)g(t)\,dt + C\right]$$

(exponential)

Uniqueness and existence:

If p and q are continuous functions on open interval $I = (\alpha, \beta)$ containing point $t = t_0$, then there exists a unique function $y = \phi(t)$ that satisfies the diff. eqn $y' + p(t)y = g(t)$ for each t in I and satisfies the initial condition $y(t_0) = y_0$ where y_0 is an arbitrary prescribed initial value.

Theorem: let functions $\overset{\text{existence}}{f}$ and $\overset{\text{uniqueness}}{\frac{\partial f}{\partial y}}$ be continuous in some rectangle $\alpha < t < \beta$, $\gamma < y < \delta$ containing the point (t_0, y_0). Then, in some interval $t_0 - h < t < t_0 + h$ contained in $\alpha < t < \beta$, there is a unique soln. $y = \phi(t)$ of the IVP $y' = f(t,y)$, $y(t_0) = y_0$



↑ If $\frac{\partial f}{\partial y}$ is discontinuous at (t_0, y_0), then Existence - Uniqueness does not apply

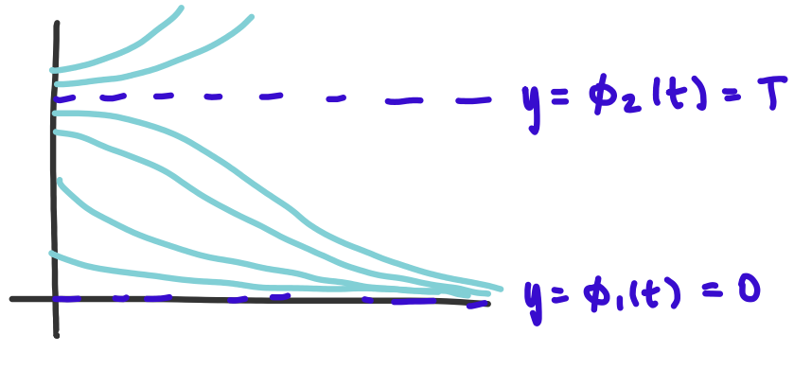
Logistic equation: $\frac{dy}{dt} = (r - ay)y$

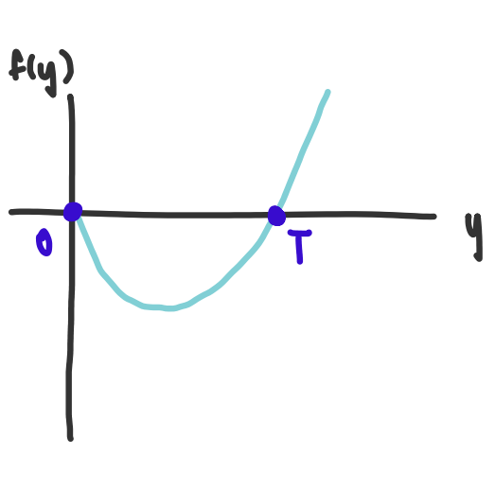
$$= r\left(1 - \frac{y}{K}\right)y$$

r: intrinsic growth rate

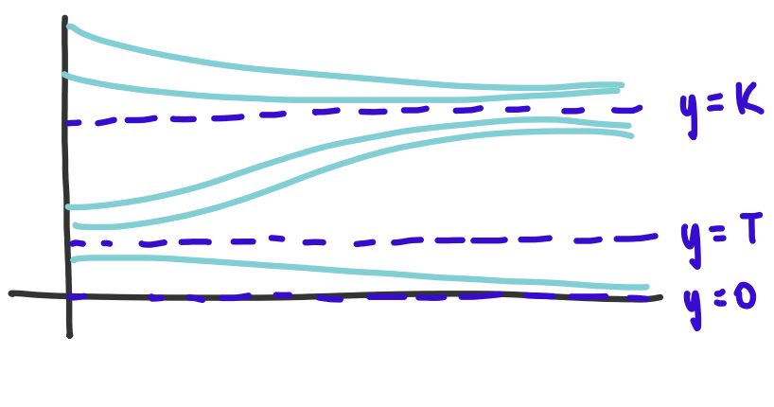
$K = \frac{r}{a}$: carrying capacity

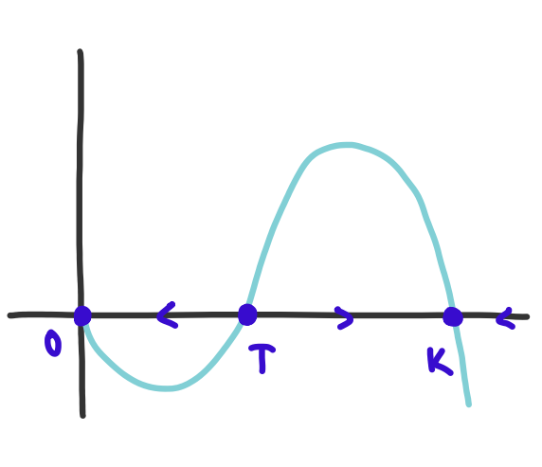
(1) $\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)y$





(2) $\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)\left(1 - \frac{y}{K}\right)y$, $r > 0$, $0 < T < K$





Homogeneous linear systems

$$X'(t) = AX(t)$$

If $x_1(t)$, $x_2(t)$ are solns, then $X(t) = c_1x_1(t) + c_2x_2(t)$ is a soln.

Wronskian: $W[x_1, x_2](t) = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}$ ≠0 if unique soln.

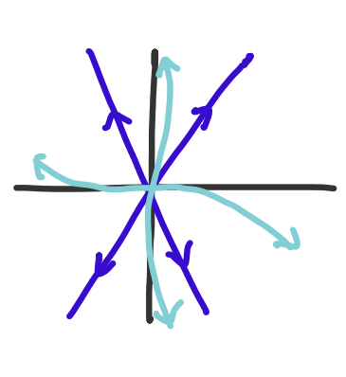
(↑ x_1, ↑ x_2)

x_1, x_2 are a fundamental set if LI.

If λ is an eigenvalue and v is corresponding eigenvector, $x = e^{\lambda t}v$ is a solution of $\frac{dx}{dt} = Ax$.

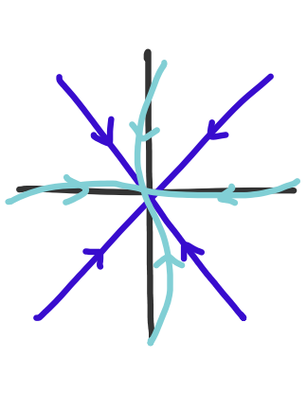
$\lambda_1 \neq \lambda_2 > 0$

nodal source



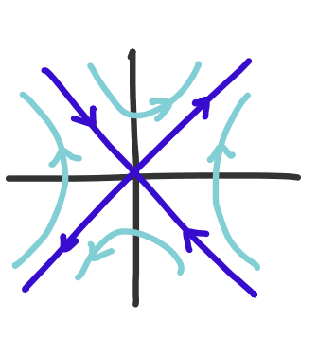
$\lambda_1 \neq \lambda_2 < 0$

nodal sink



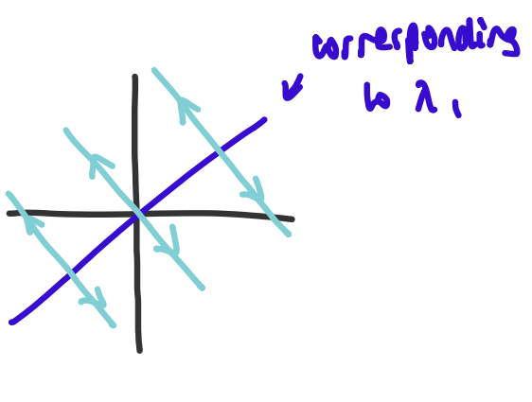
$\lambda_1 < 0 < \lambda_2$

saddle



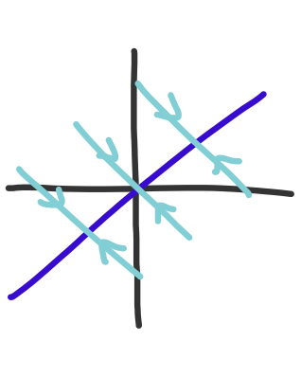
$\lambda_1 = 0$, $\lambda_2 > 0$

line of nodes



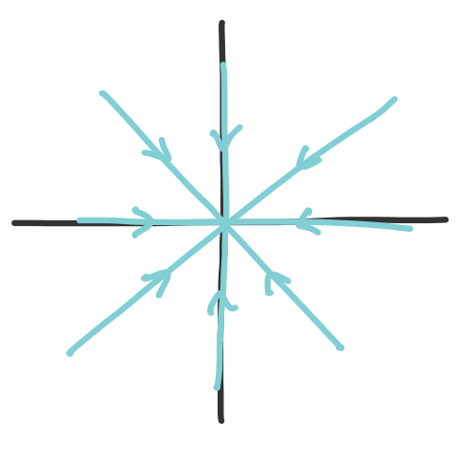
$\lambda_1 = 0$, $\lambda_2 < 0$

line of nodes



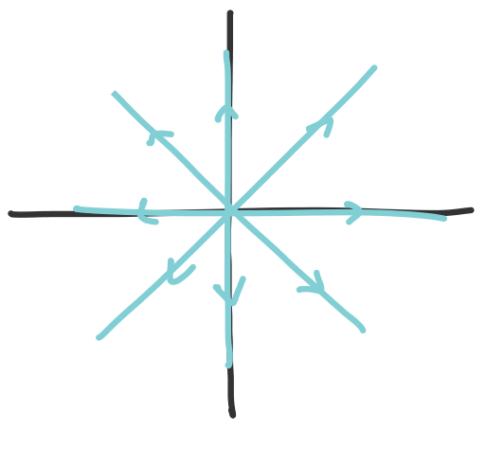
$\lambda_1 = \lambda_2 < 0$, 2 distinct eigenvectors

Stable star node



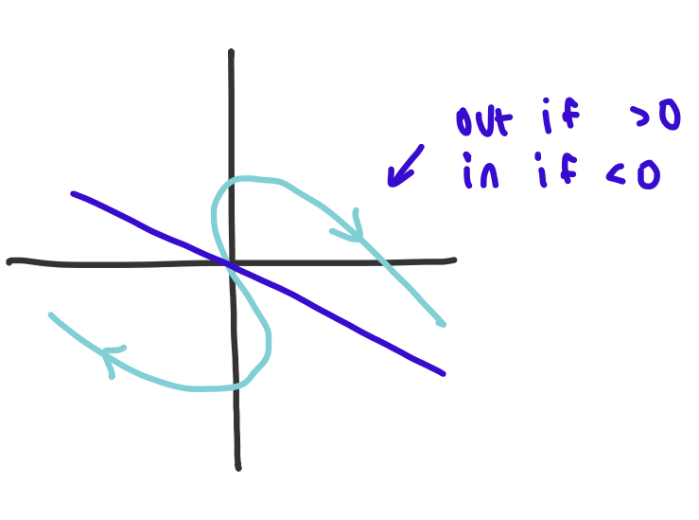
$\lambda_1 = \lambda_2 > 0$, 2 distinct eigenvectors

unstable star node



$\lambda_1 = \lambda_2$, 1 distinct eigenvector

degenerate node



Complex eigenvalues

complex eigenvalues come in conjugate pairs

① find eigenvalues $\lambda = \mu \pm i\nu$

② determine eigenvector corresponding to $\lambda_1 = \mu + i\nu$ by solving $(A - \lambda_1 I)v = 0$

③ express eigenvector in form $a + bi$

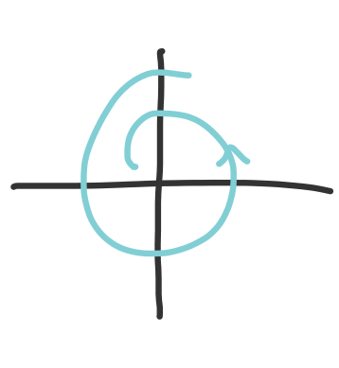
④ write soln $x_1(t) = \underbrace{e^{\mu t}(a\cos\nu t - b\sin\nu t)}_{u(t)} + \underbrace{ie^{\mu t}(a\sin\nu t + b\cos\nu t)}_{w(t)}$

(fundamental set)

⑤ general soln: $x(t) = c_1u(t) + c_2w(t)$

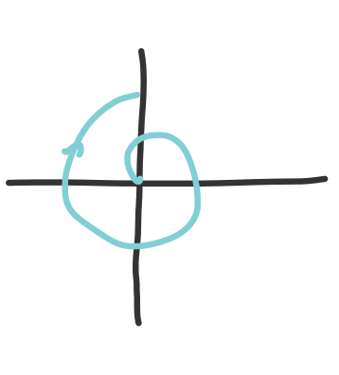
$\lambda = \mu \pm i\nu$, $\mu < 0$

spiral sink



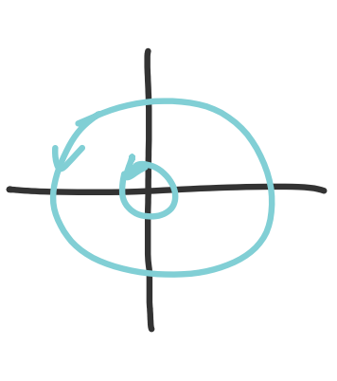
$\lambda = \mu \pm i\nu$, $\mu > 0$

spiral source



$\lambda = \mu \pm i\nu$, $\mu = 0$

center



Repeated eigenvalues

$$x_1(t) = e^{\lambda t}v_1$$

second soln: assume $x = te^{\lambda t}v + e^{\lambda t}w$

find v, w by equating coefficients

Second order Linear Equations

$y'' + p(t)y' + q(t)y = g(t)$ p, q, g continuous on I

let $x = \begin{pmatrix} y \\ y' \end{pmatrix}$

$$x' = \begin{pmatrix} 0 & 1 \\ -q(t) & p(t) \end{pmatrix}x + \begin{pmatrix} 0 \\ g(t) \end{pmatrix}$$

Two solns y_1 and y_2 to homogeneous eqn. $y'' + p(t)y' + q(t)y = 0$ form a fundamental set if $W[y_1, y_2] = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0$

If $x_1, \dots x_n$ LI, then $W[x_1, \dots x_n](t) \neq 0$ for every point

If $x_1 \dots x_n$ not LI, then $W[x_1, \dots x_n](t) = 0$ for every point

Corollary: If functions $p_1(t), p_2(t), \dots p_n(t)$ and $g(t)$ are continuous on open interval $I = (\alpha, \beta)$, then there exists exactly one soln. $y = \phi(t)$ of the diff. eqn.

$$\frac{d^ny}{dt^n} + p_1(t)\frac{d^{n-1}y}{dt^{n-1}} + \dots + p_{n-1}(t)\frac{dy}{dt} + p_n(t)y = g(t)$$

that also satisfies initial conditions $y(t_0) = y_0$, $y'(t_0) = y_1$, ... $y^{(n-1)}(t_0) = y_{n-1}$

and it exists throughout I.

Constant coefficients: $ay'' + by' + c = 0$ characteristic eqn: $a\lambda^2 + b\lambda + c = 0$

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Real and distinct roots: $y = c_1e^{\lambda_1 t} + c_2e^{\lambda_2 t}$

Repeated roots: $\lambda_1 = \lambda_2 = -\frac{b}{2a}$

$y = c_1e^{\lambda t} + c_2te^{\lambda t}$

Complex roots: $\lambda_1 = \mu + i\nu$, $\lambda_2 = \mu - i\nu$

$y = c_1e^{\mu t}\cos\nu t + c_2e^{\mu t}\sin\nu t$

If x_p is a particular soln to $x' = P(t)x + g(t)$ and $\{x_1, x_2\}$ is a fundamental set of solutions to the corresponding homogeneous eqn,

general soln: $x = c_1x_1(t) + c_2x_2(t) + \underline{x_p(t)}$

If Y is a particular soln to $y'' + p(t)y' + q(t)y = g(t)$ and $\{y_1, y_2\}$ is a fundamental set of solutions corresponding to homogeneous eqn,

general soln: $y = c_1y_1(t) + c_2y_2(t) + \underline{Y(t)}$

Method of undetermined coefficients

$P_n(t) = A_0t^n + A_1t^{n-1} + \dots + A_n \rightarrow t^s(A_0t^n + A_1t^{n-1} + \dots + A_n)$

$P_n(t)e^{\alpha t} \rightarrow t^s(A_0t^n + A_1t^{n-1} + \dots + A_n)e^{\alpha t}$

$P_n(t)e^{\alpha t}\sin\beta t$ (or $\cos\beta t$) $\rightarrow t^s[(A_0t^n + A_1t^{n-1} + \dots + A_n)e^{\alpha t}\cos\beta t + (B_0t^n + B_1t^{n-1} + \dots + B_n)e^{\alpha t}\sin\beta t]$

s is smallest integer such that no term in Y_i is a solution of homogeneous eqn.

Variation of parameters

If x_1, x_2 form fundamental set of $x' = P(t)x$, then a particular soln to $x' = P(t)x + g(t)$ is $x_p(t) = X(t)\int X^{-1}(t)g(t)\,dt$, where $X(t) = [x_1(t), x_2(t)]$

If $y'' + p(t)y' + q(t)y = g(t)$,

$$Y(t) = -y_1(t)\int \frac{y_2(t)g(t)}{W[y_1, y_2](t)}\,dt + y_2(t)\int \frac{y_1(t)g(t)}{W[y_1, y_2](t)}\,dt$$

where y_1, y_2 form fundamental set to homogeneous eqn.

Matrix exponential: $e^{At} = X(t)X^{-1}(0)$ $\left(\vec{x}(t) = e^{At}\vec{x}(0)\right)$

$$e^{\lambda t}\begin{pmatrix} c_1 + \underset{\uparrow}{tc_2} \\ c_2 \end{pmatrix}$$

will dominate as $t \to \infty$ (ie. degenerate node)

$$e^{At} = \sum_{n=0}^{\infty} \frac{A^nt^n}{n!}$$

$x' = P(t)x + g(t)$

↙ for first order linear systems

Existence + uniqueness: If $P(t)$ and $g(t)$ are continuous on $I = (\alpha, \beta)$, then there exists a unique soln. $x = \phi(t)$ of the IVP $x' = P(t)x + g(t)$, $x(t_0) = x_0$, where $t_0 \in I$

If an autonomous system, given any values of x and y, there is exactly one soln passing through that point regardless of time. (applies to whole system, not just one variable)

Existence & uniqueness

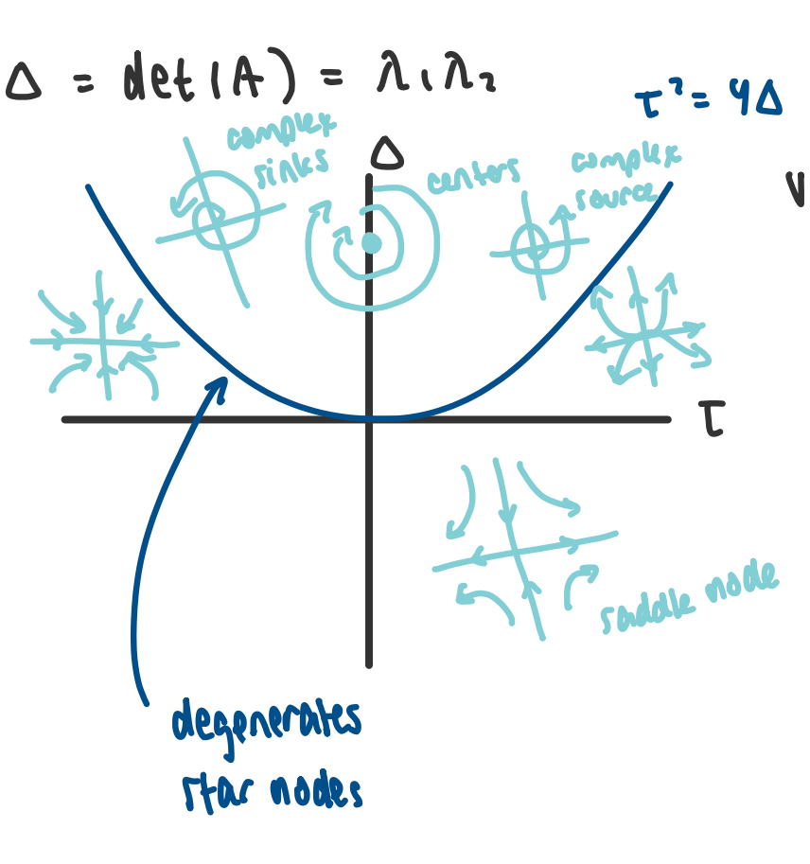
Linear:	$y' + P(t)y = Q(t)$	$P(t)$, $Q(t)$ continuous
Nonlinear:	$y' = f(t,y)$	f, $\frac{\partial f}{\partial y}$ continuous
System:	$X' = A(t)X + F(t)$	$A(t)$, $F(t)$ continuous

Trace determinant plane

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$\tau = Tr(A) = \lambda_1 + \lambda_2$

$\Delta = \det(A) = \lambda_1\lambda_2$



$v_1e^{\lambda_1 t} + v_2e^{\lambda_2 t}$

$\lambda = -1+i, -1-i$

$$x_1 = e^{(-1+i)t}\begin{pmatrix} 2+i \\ 1 \end{pmatrix}$$

$$= e^{-t}e^{it}\begin{pmatrix} 2+i \\ 1 \end{pmatrix}$$

$$= e^{-t}(\cos t + i\sin t)\begin{pmatrix} 2+i \\ 1 \end{pmatrix}$$

$$= e^{-t}\left[\underbrace{\begin{pmatrix} 2\cos t - \sin t \\ \cos t \end{pmatrix}}_{u} + i\underbrace{\begin{pmatrix} 2\sin t + \cos t \\ \sin t \end{pmatrix}}_{v}\right]$$

$W[u\ v] \neq 0$

$$x(t) = c_1e^{-t}\begin{pmatrix} 2\cos t - \sin t \\ \cos t \end{pmatrix} + c_2e^{-t}\begin{pmatrix} 2\sin t + \cos t \\ \sin t \end{pmatrix}$$

In general, if there is dependence on both x and t, and terms are not linear, then it is not separable